

Tanuj Dargan

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EDUCATION & AWARDS

University of Victoria – International Scholarship

B.S. Computer Science – Accelerated 3-Year Completion

Relevant Coursework: Quantum Computing, Computer Architecture, Numerical Analysis, Communications & Networks

Victoria, BC

Apr. 2027

WORK EXPERIENCE

Amazon Robotics

May 2026 – Aug. 2026

Software Development Engineer Intern

Toronto, ON

- Built and **deployed an LLM-powered planning assistant to production** within Optimus, Amazon Robotics' daily labor optimization platform serving **3,000+ fulfillment sites across NA and EU**, enabling site planners to query plans, surface anomalies, and generate labor recommendations via **natural language** instead of manual dashboard navigation.
- Implemented a **RAG pipeline** over historical labor plans, site configuration docs, and operational constraints, backed by **vector search** and prompt-engineered chains that contextualize responses to site-specific parameters (ARS, TNS, and more), **reducing planner time-to-insight by 60%+**.
- Designed a **multi-turn conversational interface** that integrates directly with Optimus' optimization engine, allowing planners to simulate **what-if labor scenarios**, adjust constraints in natural language, and receive **real-time re-optimized plans** without re-running the full solver pipeline.

RBC Borealis – AI Mentorship Program

Mar. 2026 – May 2026

AI/ML Research Mentee

Remote

- Conducted exploratory data analysis on **1,500+ longitudinal youth health engagement records** spanning 3+ years in a team of 4 mentees alongside 2 senior AI researchers, surfacing **disengagement signals across 20+ behavioural features** using Python, Pandas, and NumPy.
- Reduced high-dimensional engagement data to **2D embeddings via UMAP clustering** in Python and scikit-learn, isolating **4+ distinct user cohorts** to inform downstream model design.
- Engineered an **ML pipeline to flag at-risk youth** navigating chronic illness by training **PyTorch models on 3+ years of longitudinal engagement data**, enabling **earlier clinical intervention**.
- Designed an analytics dashboard serving **4+ clinical and program teams** by visualizing **retention curves, engagement trends, and disengagement risk flags** using Python, Matplotlib, and GA4.

Pear Care

Jul. 2025 – Present

Lead AI Developer

Remote

- Accelerated diagnostic workflows by building **autonomous medical-AI agents** with LangGraph, **MoE routing**, and task-introspective reasoning, served via FastAPI on GCP across **1,000+ clinical specialties**.
- Cut **model memory footprint by 40%** while preserving **95%+ F1** on internal clinical triage benchmarks by self-hosting with PyTorch, Ollama, and vLLM, then transitioning to **quantized LoRA/PEFT** fine-tuned deployments.
- Architected for **horizontal scale** across a multi-tenant **GCP/AWS infrastructure** with Redis caching, DynamoDB persistence, **end-to-end encryption**, and mobile-first access, validated through load and stress testing.

Major League Hacking (MLH) SWE Fellowship - Apache Airflow

May 2025 – Aug. 2025

Software Engineering Fellow

Remote

- Streamlined developer debugging for Apache Airflow's **50K+ star codebase** by architecting **Docker-based IDE integration** (VSCode, PyCharm) with step-through debugging and dynamic port exposure, cutting **setup-to-debug cycles from hours to minutes**; shipped in a team of 3 alongside Royal Bank of Canada engineers.
- Diagnosed and patched a **long-standing time-window logic bug** causing **silently missed DAG runs** across multiple Airflow releases; fix **merged into the core scheduler**.

DEIA (Database & Data Mining)

May 2025 – Present

Undergraduate Researcher - Prof. Sean Chester

Victoria, BC

- Built a custom retrieval system on a modified cuVS fork (CAGRA, IVF-PQ) sustaining **1–4M QPS at billion-vector scale** with **90–95% recall** and **sub-250 ms latency** across 8+ NVIDIA GPUs and 2TB of indexed data.
- Achieved **4.3× throughput scaling** by building automated stress-testing pipelines on **Kubernetes GPU clusters**, profiling memory bandwidth and **NVLink interconnect saturation** to expose bottlenecks.

UVic Centre for Aerospace Research (CFAR)

Apr. 2025 – Present

Software Developer & Undergraduate Researcher

Victoria, BC

- Led **OBC firmware (C)**, **TTC board design**, and **cross-functional hardware/software integration** for DoomSat and a high-altitude balloon mission, coordinating across electrical, mechanical, and software teams.
- Cut MarmotSat testing from **180 min to 15 min** by building a **CI/HIL pipeline** with GitHub Actions and SWD flashing (80+ checksum rules/commit), and implementing C/Java YAMCS integration that eliminated **20+ hours of manual work per release**.

PROJECTS

- Hive Agent Framework – Open Source Contributor** | *7.7K+ Stars, Python, Bash* Jan. 2026 – Present
- Resolved a **cross-platform setup blocker** (Issue #476) by overhauling Python detection with **multi-candidate probing**, timeout handling, and **pipefail** hardening across **Windows, macOS, and Linux**; actively contributing additional tooling.
- Optimate: Hack the North 2025 Winner** | *Next.js, Cohere, AWS DynamoDB* Sep. 2025
- Won **2 prizes out of 300+ teams** (**YC Unicorn Prize & Federato Best RiskOps**) by building an AI underwriting dashboard that automated risk assessment with Next.js, Cohere LLMs with RL, and DynamoDB; Federato CEO validated **40% faster review cycles** and **15% fewer underwriting errors** versus manual workflows.
- AIMO Progress Prize 2 Competitor: Silver Medal** | *Finishing score: 27* Nov. 2024 – Mar. 2025
- Achieved **54% accuracy**, finishing **top 10% of 2,200 teams** by engineering **multi-stage LLM pipelines** with advanced prompt engineering, modular reasoning strategies, and hyper-parameter tuning.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript/TypeScript, Go, Bash, SQL
AI/ML: PyTorch, vLLM, Ollama, LangGraph, LoRA/PEFT, Hugging Face, OpenCV, Pandas, cuVS
Frameworks & Tools: FastAPI, Next.js, React, Node.js, Flutter, Django, Flask, Docker, Kubernetes, Git, GitHub Actions
Infrastructure: AWS (EC2, S3, Lambda, SageMaker, DynamoDB, Bedrock), GCP, Redis, Proxmox, Nginx, Cloudflare, Slurm